A 1.1.2 The GPU

A 1.1.2 Describe the role of a GPU

A graphics processing unit (GPU) is a co-processor: a specialist that sits alongside the CPU and takes on the work it is built to do faster. Its speciality is applying the same operation to enormous numbers of data items at the same time. That one idea explains both its original job, rendering graphics, and its newer ones, from training neural networks to running large scientific simulations. This page describes the GPU's role, the architecture that suits it to this kind of work, and the real-world scenarios the syllabus names.

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Key Questions

- What is the role of a GPU in a computer system?
- Which features of GPU architecture make it suitable for complex computations?
- Why do video games, AI, machine learning and large simulations all benefit from the same processor design?

IB Command Term note

DESCRIBE: Give a detailed account. State what the GPU does, name the architectural features that make it suitable (many simple cores, parallel execution, high-bandwidth dedicated memory), and connect each application to those features.

WHAT TO INCLUDE: The GPU's role as a co-processor; its architecture; and real-world scenarios: video games, AI, large simulations, graphics rendering and machine learning.

The role of a GPU

A 1.1.1 introduced co-processors: specialists that take particular work off the main CPU. The GPU is the most important of them. Where the CPU handles varied tasks one after another, the GPU is designed to process large volumes of data where the same operation must be applied to many data points at once.

Graphics rendering remains the core function, and it is why the architecture exists. Typical tasks include 3D modelling (generating and rendering complex scenes for games, simulations and animation), image processing (applying filters, adjusting colour, enhancing visual output) and video encoding and decoding (compressing and decompressing video for playback and streaming).

General-purpose GPU computing (GPGPU) extends the same capability to non-graphical work that benefits from parallelism: scientific computing (climate modelling, drug discovery), large-scale data analysis, machine learning (training neural networks), and cryptocurrency mining (repeated hash calculations). None of these is graphics, yet all of them are the same shape of problem: one operation, millions of data points.

The architecture that makes it suitable

Three architectural features explain the GPU's effectiveness, and a Describe answer should be able to name all three.

- **Many simple cores.** A GPU contains hundreds or thousands of cores. Each is far simpler than a CPU core: it cannot branch or make complex decisions. The cores work in lockstep, all performing the same task at the same moment.
- **Parallel execution.** One instruction is applied to many data items simultaneously, like applying a filter to every pixel of an image in one pass rather than pixel by pixel. The diagram below shows the principle.
- **High-bandwidth dedicated memory.** Thousands of cores reach for data at once, so the GPU has its own wide, fast memory to keep them all supplied without a bottleneck.

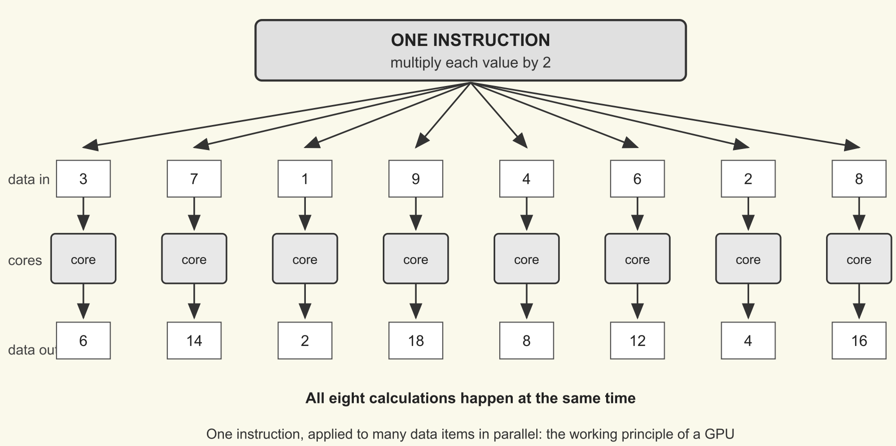

Figure 1. One instruction, applied to many data items in parallel: the working principle of a GPU.

COMMON EXAM MISTAKE

Describing the GPU as a faster CPU, or as the chip that makes games run. A GPU is not faster in general; it is faster only at parallel work, and weaker than a CPU at varied sequential work. A Describe answer should link the architecture (many simple cores, parallel execution, dedicated memory) to the task, not claim general superiority.

Real-world scenarios

Video games. A GPU processes millions of pixels per frame in real time, handling lighting calculations, texture mapping and effects simultaneously. That parallelism is what makes detailed, high-frame-rate rendering possible.

AI and machine learning. Training a neural network involves enormous numbers of matrix operations, which break down into independent calculations that can run in parallel. GPUs cut training times from weeks to days or hours, which is what made today's large models practical.

Large simulations. A climate model or a drug-discovery simulation applies the same physics or chemistry calculation to millions of grid cells or molecules. Each cell is independent, so the work maps directly onto thousands of cores.

EXAM TIP

When a question names an application, structure the answer in two parts: first identify the parallel workload (millions of pixels, matrix operations, grid cells), then name the architectural feature that handles it. Linking the application to the architecture is what the command term **Describe** is asking for here.

Summary

SUMMARY REFERENCE

ASPECT	WHAT TO SAY
Role	A co-processor specialised in applying the same operation to many data points at once; graphics rendering plus GPGPU.

ASPECT	WHAT TO SAY
Architecture	Hundreds or thousands of simple cores working in lockstep; parallel execution of one instruction over many data items; high-bandwidth dedicated memory.
Scenarios	Video games (pixels, lighting, textures per frame); AI and machine learning (parallel matrix operations); large simulations (same calculation over millions of cells).

Try it on paper

The quiz below will help you test your knowledge. Answer this question to practice writing an exam style question.

Describe how the architecture of a GPU makes it suitable for training a machine learning model. [3 marks]

Model answer: *Training involves vast numbers of matrix operations that can be split into independent calculations. [1] The GPU's hundreds or thousands of simple cores perform these calculations in parallel rather than one at a time. [1] Its high-bandwidth dedicated memory keeps all the cores supplied with data, so training completes in a fraction of the time a CPU would need. [1]*

A 1.1.2 Quiz

1

What is the PRIMARY function of a GPU?

A. To connect the computer to the internet.

B. To manage the operating system and applications.

C. To accelerate the creation of images and videos.

D. To store large amounts of data.

2

Which of the following tasks is NOT typically handled by a GPU?

A. Running a word processing application.

B. Rendering 3D graphics in a video game.

C. Training a machine learning model.

D. Applying filters to an image.

3

What architectural feature allows GPUs to excel at parallel processing?

A. A massive number of cores.

B. A complex control flow mechanism.

C. A large amount of cache memory.

D. A single, powerful core.

4

How does the memory organization of a GPU differ from a CPU?

A. GPUs have slower memory than CPUs.

B. GPU memory is primarily used for storing programs.

C. GPU memory is optimized for parallel access by many cores.

D. GPUs have less memory than CPUs.

5

What is GPGPU?

A. The use of GPUs for general-purpose computing tasks.

B. A programming language for GPUs.

C. A specialised type of GPU for scientific simulations.

D. A type of graphics rendering technique.

6

Why are GPUs well-suited for AI and machine learning tasks?

A. They have built-in AI algorithms.

B. They can access and process data from the internet faster.

C. They have specialised cores for artificial intelligence.

D. They can efficiently handle the matrix operations common in AI.

7

Which of the following is NOT a common GPU programming model?

A. Python

B. CUDA

C. OpenGL

D. OpenCL

8

What is a key advantage of using GPUs for scientific simulations?

A. They can handle complex calculations faster than CPUs.

B. They consume less power than CPUs.

C. They have built-in physics engines.

D. They are easier to program than CPUs.

9

How do GPUs contribute to the advancement of AI?

A. By enabling AI to communicate with humans.

B. By automatically generating AI algorithms.

C. By providing faster processing for training AI models.

D. By replacing the need for large datasets.

10

What is a potential future trend for GPU technology?

A. GPUs will be replaced by FPGAs for all applications.

B. GPUs will no longer be used for graphics rendering.

C. GPUs will have even more cores and specialized hardware for AI.

D. GPUs will become less specialized and more like CPUs.

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